

Micro I, class 10

Last time we proved a monotone comparative statics theorem (attributed to Milgrom-Shannon, 1994): if a function f on $X \times T$, $X \subset \mathbb{R}$, $T \subset \mathbb{R}$ satisfies the *strict single crossing property* (SSCP) in (x, t) , then $f(\cdot, t)$ -maximizing choices of x *do not decrease* with an increase in t . Today I took up the more interesting and applicable case in which both x and t are *vectors*, not scalars. The definition of the SSCP is the *same*, word for word, as in the scalar case. But the conclusion does not follow simply from the SSCP.

To begin, I introduced a stronger and often-easier-to-check condition, *strictly increasing differences* in x and t : for every $x'' > x'$, $t'' > t'$, $f(x'', t) - f(x', t)$ is strictly increasing in t , or

$$f(x'', t'') - f(x', t'') > f(x'', t') - f(x', t').$$

Topkis's Theorem (1978) asserts that if f satisfies strictly increasing differences in x and t , and is supermodular (SPM) on X for each $t \in T$, then $f(\cdot, t)$ -maximizing choices of x *do not decrease* with an increase in t . (Formally, if f satisfies SID in (x, t) in t^0, t^1 in T and $f(\cdot, t)$ is SPM for $t^1 > t^0$, and $x^i \in \operatorname{argmax}_{x \in X} f(x, t^i)$, then $x^1 \geq x^0$.) The SPM condition says, loosely, that the choice variables are themselves complementary to each other: an increase in x_i increases the net marginal benefit of an increase in x_j . (Indeed as I point out in my notes, SPM of $f(\cdot, t)$ on X is equivalent to pairwise ID for all possible comparisons between the components of the choice vector).

Topkis's theorem leads to a simple (but strong) *sufficient* condition for monotone comparative statics if f is C^2 :

$$\frac{\partial^2 f(y)}{\partial y_i \partial y_j} > 0$$

for every pair of variables y_i and y_j , $i \neq j$, that enter the function f (no matter whether they are choice variables or parameters).

We applied these theorems to the problem of a firm with a single-output technology to the question of how an increase in the output price affects the demand for the inputs: if the production function f is strictly increasing, then profit satisfies SID in p and z ; and if f is SPM, then so is profit. So, if f is strictly increasing and SPM, the demand for each input *does not decrease* when the output price *increases*.

After finishing up our discussion of comparative statics—in particular Milgrom-Shannon's Theorem 4'—I will briefly consider cost functions and some of their properties. The last topic will be competitive equilibria and Pareto optimality in the setting a simple general equilibrium model, the *two-good quasilinear economy*. It gives a foundation of many of the models used in applied microeconomics. I have extensive notes on this model on Canvas.